

Improved Privacy

Laryngectomy patient

Multiple Sclerosis Dysphonia

Robustness to environmental noises

Restricted situations

Importance [3][41]



Design for Photodiode

Nylon
Rubber casting
Elastomer band

Materials

Neck Gaiter

Flexible PCB

Ansys simulation

Low-power battery

Heat minimisation

Wearability

Computer Aided Design

FEA Simulation

Cycle tests

Durability

Biometrics Collection

User Satisfaction Survey

Ergonomics



Clocked frequency

High-density recording

680nm & 850nm light

Blender for light intensity test

LED & Photodiodes [39]

Design for camera platform

Camera [20] [22][27][57]

Sensors

NIR- 850nm

Frame rate

Illumination

FOV

IOS

Minimal focusing distance

Paramelers

System Design

Software Engineering

Downloadable Application

Compatibility

Containing

Online Meeting

Smart Furniture

Edge computing

Containing

MPU Controller

Increased heating

Increased power consumption

Faster response

Cloud computing

Containing

Require Data Transmission (Bluetooth, WiFi)

Higher data throughput

Bias

Heart rate

Ambient Light

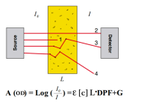
Skintone

Breathe

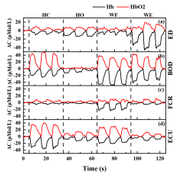
Sickness(e.g. Inflammation)

Motion

Modified Beer-Lambert Law



$$A(\alpha\beta) = \mathrm{Log}\left(\frac{I_0}{I}\right) = \varepsilon\,[c]\,L'DPF + G$$



Monte Carlo Simulation [6] [26]

Modified B-L Law [25]

Beer-Lambert Law

NIR Pathway in Tissues [49]

Haemoglobin (Hb) [34][40][58]

Oximetry [16]

Myoglobin (Mb) [30]

Biophotonics [29][43]

Near-Infrared Silent Speech Interface System [8][19][32]

Physiology

Phonation [11][15]

Swallowing and respiration [53]

Multi-layered Tissue [26]

Cricoarytenoid

Thyroarytenoid

Cricothyroid

Sternohyoid

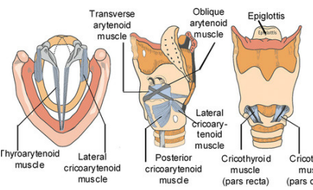
Thyrohyoid muscle

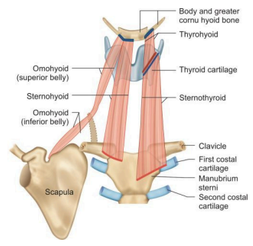
Epidermis

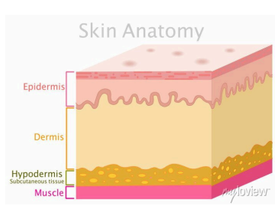
Dermis

Adipose Tissue

Muscle







Analysis Pipeline

Signal Processing [21]

Dimensionality Reduction

Containing

[47] Active Appearance Model

[15][53] ICA

[10] PCA

[23] 2D-2PCA

[33][18] "Eigenface"

Sampling Window

Containing

Fixed moving

Adaptive/variable

Filtering

Containing

Batterworth Filter

Notch Filter

Mel-spectrogram

Discrete Cosine Transform [19]

Common average removal [15][53]

Dimensionality Reduction

Tailored Dataset

Corpus

Containing

VCTK

LibriSpeech

Deeplake

LJ Speech

Common Voice

Phoneme boundary identification

Containing

Montreal Forced Aligner [28]

Acoustic Model

Pronunciation dictionary

TextGrid analysis

Phoneme synthesis

Containing

Time interval extraction

Audio segmentation

Filtering

Feed to train language model

Machine Learning [7],[45]

For images

Classification [12][42]

Containing

MLP [12]

HMM [58]

DNN [12][54]

CNN [1][13][7][42][45][46]

RNN [58][59]

CRNN [41]

For speech

Regression [1][12]

Containing

RNN[2]

LSTM [58]

Self-Attention

Transformer (LLM) [14]

Voice Synthesis [14]

Encoder/Decoder[42]

For optimization

Model Generalization

Aim for

All Accents
All Languages
All Genders

Model Training Basic Formula

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(\underbrace{\hat{y}^{(i)} - y^{(i)}}_{\text{error}}\right)^2$$

m = number of training examples

Cost Function

$$tmp_w = w - \alpha\frac{\partial}{\partial w}J(w,b)$$

$$tmp_b = b - \alpha\frac{\partial}{\partial b}J(w,b)$$

$$w = tmp_w$$

$$b = tmp_b$$

Gradient Decent

$$f_{w,b}(\vec{x}) = g(\underbrace{\vec{w}\cdot\vec{x} + b}_{z}) = \frac{1}{1+e^{-(\vec{w}\cdot\vec{x}+b)}}$$

Logistic regression

Alternative SSI Paradigms

Optical Camera [36] [41][48][51]

Impulse Radio Ultra-Wideband [44]

Distance Camera [56]

Ultrasound [37] [48]

Mechanical Sensor [50]

Containing

Strain sensors [13][14]

Mangetoelasticity [60]

Optical Fibre [54]

Triboelectric [38]

EXG [52]

Containing

EMG [3][4][5][17][23][55]

EPG [34]

EEG [8][19]

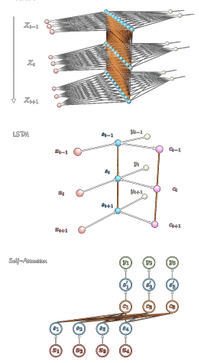
ECOG [19]

functional-NIR [31]

CNN



RNN



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Images and Use of Generative AI

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